

Question ID: 2554b413

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question

In the xy-plane, a line has a slope of 6 and passes through the point (0,8). Which of the following is an equation of this line?

Answer

- A. $y = 6x + 8$
- B. $y = 6x + 48$
- C. $y = 8x + 6$
- D. $y = 8x + 48$

Correct Answer: A

Rationale

Choice A is correct. The slope-intercept form of an equation for a line is $y = mx + b$, where m is the slope of the line and b is the y-coordinate of the y-intercept of the line. It's given that the slope is 6, so $m = 6$. It's also given that the line passes through the point (0,8) on the y-axis, so $b = 8$. Substituting $m = 6$ and $b = 8$ into the equation $y = mx + b$ gives $y = 6x + 8$.

Choices B, C, and D are incorrect and may result from misinterpreting the slope-intercept form of an equation of a line.